

VEIHOLMEN, Norway

WMO: 012250

Lat: 63.517N

Lon: 7.950E

Elev: 5

StdP: 101.26

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-5.3	-3.1	-10.9	1.5	-3.3	-9.3	1.7	-2.1	20.9	5.5	19.0	5.3	6.3	60	0.821

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.8	19.1	14.9	17.6	14.3	16.6	13.9	15.7	18.0	14.9	17.0	14.3	16.2	4.6	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.6	10.4	16.9	13.9	9.9	16.1	13.2	9.4	15.5	43.8	18.3	41.9	17.1	40.1	16.3	19.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.5	17.4	15.5	DB	-5.5	22.9	2.7	1.9	-7.4	24.3	-8.9	25.5	-10.5	26.6	-12.4	28.0	(4)
(5)			WB	-7.1	16.7	2.4	1.1	-8.8	17.4	-10.1	18.0	-11.5	18.6	-13.2	19.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.9	3.2	3.0	3.5	5.7	8.3	10.8	13.4	14.1	12.5	9.1	6.0	4.3	(6)	
	DBStd	4.64	2.57	2.78	2.48	2.28	2.39	1.95	2.16	1.77	1.93	2.43	2.87	2.94	(7)	
	HDD10.0	1162	212	197	203	130	64	11	0	0	3	44	121	177	(8)	
	HDD18.3	3826	471	430	461	378	311	224	153	131	176	285	371	436	(9)	
	CDD10.0	380	0	0	0	2	11	37	106	128	77	18	1	0	(10)	
	CDD18.3	2	0	0	0	0	0	0	1	1	0	0	0	0	(11)	
	CDH23.3	2	0	0	0	0	0	0	2	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	6.9	7.8	7.4	7.8	6.4	5.8	6.1	5.5	5.5	7.0	7.3	7.8	8.5	(14)	
(15) Precipitation	PrecAvg	1635	172	150	144	88	80	105	107	136	153	152	152	196	(15)	
	PrecMax	1965	421	297	316	261	148	184	178	274	392	280	382	380	(16)	
	PrecMin	1105	5	9	49	24	33	35	43	43	56	33	10	51	(17)	
	PrecStd	207	95	72	72	54	27	40	35	54	81	56	100	80	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.7	8.3	9.3	14.7	17.5	18.5	22.3	21.1	18.2	16.6	11.9	10.1	(19)	
		MCWB	6.3	5.8	5.7	9.4	11.8	13.2	16.3	15.4	13.5	11.4	9.9	7.8	(20)	
	2%	DB	7.9	7.6	8.0	11.9	14.5	16.0	19.5	18.9	16.7	14.2	11.1	9.4	(21)	
		MCWB	5.6	5.7	5.6	8.1	10.5	12.7	15.6	15.1	13.2	11.3	9.0	7.5	(22)	
	5%	DB	7.4	7.0	7.4	9.8	12.6	14.4	17.6	17.5	15.8	13.1	10.3	8.8	(23)	
		MCWB	5.3	5.2	5.4	6.9	9.7	11.9	14.7	14.6	13.0	10.6	8.3	6.9	(24)	
	10%	DB	6.7	6.3	6.8	8.4	11.3	13.2	16.3	16.6	15.0	12.3	9.6	8.0	(25)	
		MCWB	4.9	4.5	5.1	6.2	8.9	11.2	14.2	14.2	12.8	10.0	7.6	5.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.2	7.0	7.1	9.4	12.2	14.2	17.4	16.5	14.7	12.6	10.5	8.7	(27)	
		MCDB	8.2	7.7	7.9	12.9	16.1	17.5	21.2	19.0	16.6	14.4	11.5	9.6	(28)	
	2%	WB	6.5	6.2	6.6	8.1	10.9	13.0	15.9	15.7	14.1	11.6	9.6	8.0	(29)	
		MCDB	7.6	7.0	7.4	10.8	13.8	15.2	18.6	17.8	15.6	13.7	10.8	9.1	(30)	
	5%	WB	5.7	5.4	5.9	7.2	10.0	12.1	15.1	15.0	13.6	11.0	8.6	7.1	(31)	
		MCDB	7.0	6.4	7.0	8.8	12.3	13.9	17.3	17.1	15.2	13.1	10.1	8.6	(32)	
	10%	WB	5.1	4.6	5.1	6.6	9.2	11.4	14.4	14.4	13.1	10.2	7.7	6.2	(33)	
		MCDB	6.6	6.0	6.6	7.9	10.8	13.0	16.4	16.3	14.8	12.2	9.4	7.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	2.9	3.0	3.2	3.2	3.1	2.7	2.8	2.8	3.0	2.9	2.9	3.1	(35)	
		MCDBR	3.1	3.2	3.7	5.7	5.9	5.2	5.8	4.9	4.1	3.7	3.0	3.6	(36)	
	5% WB	MCWBR	2.9	3.0	2.9	3.1	3.1	2.9	2.8	2.5	2.5	2.6	2.8	3.4	(37)	
		MCWBR	3.0	2.9	3.2	4.4	5.4	4.4	5.2	4.2	3.5	3.5	3.0	3.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.397	0.316	0.339	0.359	0.354	0.355	0.364	0.355	0.341	0.315	0.368	0.358	(40)	
		taud	2.661	2.262	2.338	2.379	2.456	2.485	2.487	2.512	2.520	2.411	2.596	2.469	(41)	
	Ebn at Noon		337	607	753	821	861	866	847	822	759	614	331	161	(42)	
		Edh at Noon	30	59	80	95	96	95	93	83	68	50	27	17	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.13	0.59	1.78	3.45	4.75	5.10	4.67	3.58	2.12	0.92	0.22	0.06	(44)	
	RadStd		0.02	0.10	0.27	0.31	0.21	0.51	0.42	0.34	0.25	0.14	0.04	0.01	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (30 neighbors)	+0.43	+0.67	N/A	N/A	N/A	N/A	-139	-183	+58	N/A

Nomenclature: See separate page